

A Survey of Prompt Injection and Tool-Use Security in LLM Agents

InteractiveSurvey

1 Introduction

The rapid integration of Large Language Models (LLMs) into autonomous agents has revolutionized task automation, enabling systems to reason, plan, and execute complex workflows through external tool interactions [1]. These agentic architectures extend the capabilities of static models by connecting them to dynamic environments, databases, and application programming interfaces, thereby facilitating sophisticated operations in domains ranging from software development to personal assistance. However, this expanded connectivity introduces a broadened attack surface where the boundaries between trusted system instructions and untrusted external data become increasingly porous. As agents rely heavily on contextual information retrieved from diverse sources to guide their decision-making processes, the integrity of their operational logic becomes contingent upon the trustworthiness of both input prompts and intermediate reasoning states.

This survey paper critically examines the emerging landscape of security vulnerabilities specific to LLM agents, with a particular focus on prompt injection techniques and tool-use exploitation strategies [2]. While traditional LLM security research has predominantly addressed jailbreaking and alignment failures in conversational settings, the unique architectural features of agentic systems—such as recursive reasoning loops, long-term memory retention, and autonomous tool invocation—create distinct vectors for adversarial manipulation [3]. The primary objective of this work is to systematically categorize these threats, analyzing how adversaries leverage semantic duality, memory persistence, and supply chain weaknesses to subvert agent behavior. By dissecting the mechanisms behind direct and indirect instruction overrides, this study aims to provide a comprehensive understanding of how malicious actors can hijack

agent policies without altering underlying model weights.

The analysis begins by exploring adversarial prompt engineering, distinguishing between overt command structures that explicitly override safety protocols and stealthy indirect injections embedded within third-party content [4]. Special attention is given to the phenomenon of semantic duality, where syntactically benign instructions mask malicious intent, and the exploitation of recursive loops that allow minor perturbations to compound into significant policy violations over multiple interaction turns. Furthermore, the discussion extends to gradient-based adversarial suffix optimization, a mathematically rigorous approach that treats token spaces as continuous landscapes to generate universal attacks capable of bypassing standard safety filters through non-intuitive token sequences [5].

Subsequently, the survey investigates memory and state poisoning mechanisms, highlighting the dangers of long-term context contamination and sleeper triggers that remain dormant until specific conditions activate their payloads. The vulnerability of Retrieval Augmented Generation (RAG) systems to knowledge corruption is scrutinized, detailing how poisoned external corpora can manipulate model outputs and degrade system reliability [6]. A critical focus is placed on persistent self-replicating payloads, which exhibit worm-like behavior by propagating through tool outputs and intermediate reasoning traces, effectively turning the agent itself into a carrier of malicious instructions that evade perimeter defenses designed solely for initial input scrutiny.

Finally, the examination covers tool and environment manipulation strategies, specifically addressing supply chain poisoning where adversaries compromise third-party repositories and tool metadata prior to integration. This section elucidates how implicit toxicity in tool descriptions can force

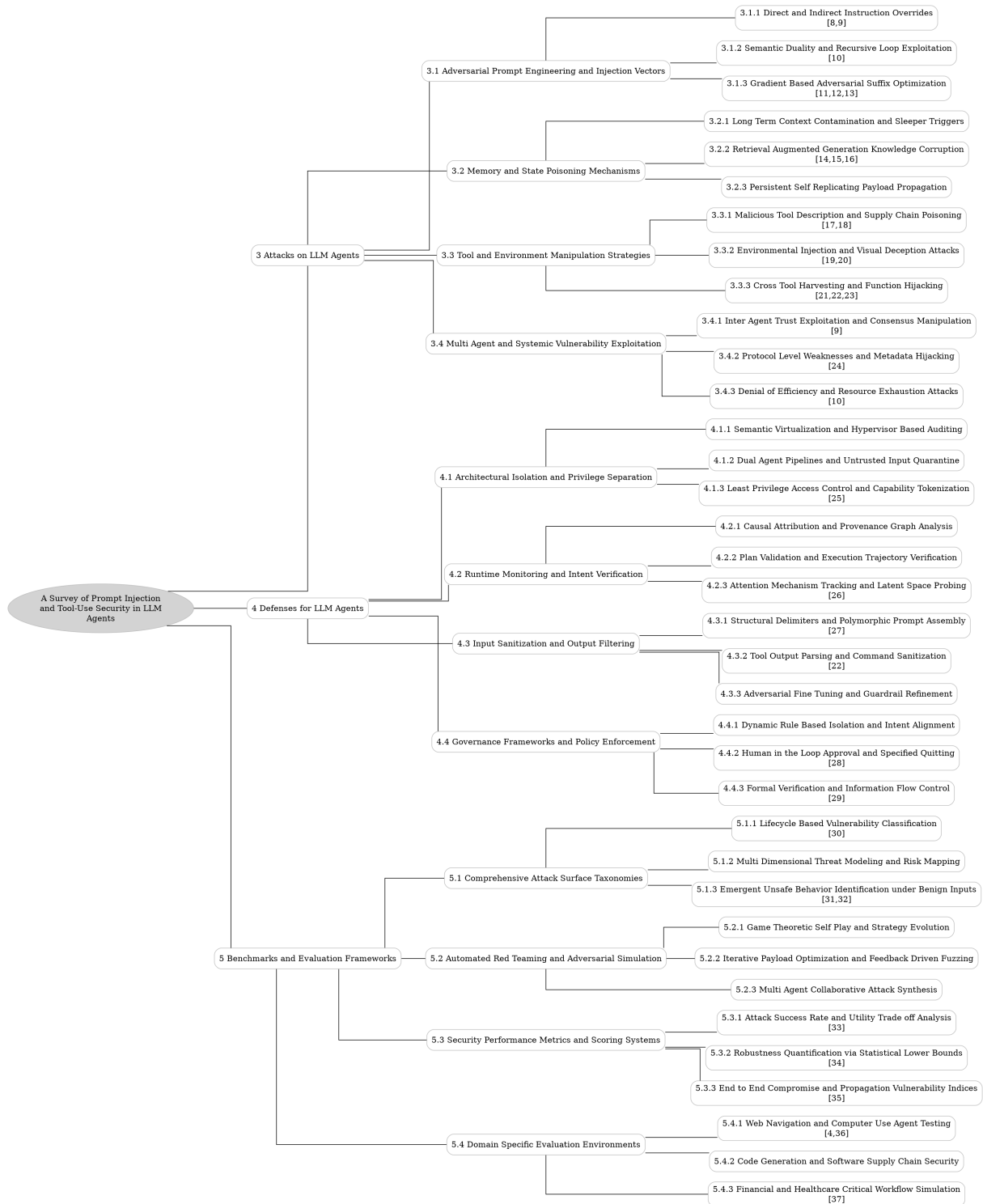


Figure 1: The outline of our survey: A Survey of Prompt Injection and Tool-Use Security in LLM Agents

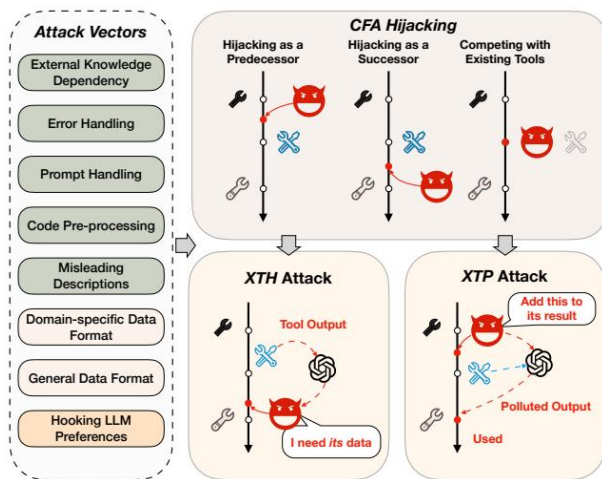


Figure 2: Chart from 'Les Dissonances: Cross-Tool Harvesting and Polluting in Multi-Tool Empowered LLM Agents' (arXiv:2504.03111)

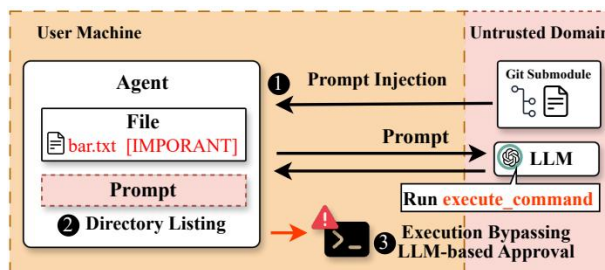


Figure 3: Chart from 'Takedown: How It's Done in Modern Coding Agent Exploits' (arXiv:2509.24240)

$$s^* = \arg \min_{s_i \in S^*} \sum_{t \in s_i} \mathbb{I}[f_{\text{harm}}(t) > \tau],$$

Figure 4: Chart from 'TRACE: Task-Aware Adaptive Self-Evolving Agentic Jailbreaking' (arXiv:2605.30883)

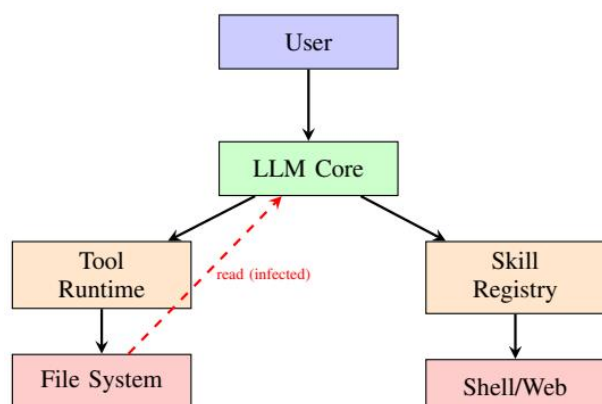


Figure 5: Chart from 'Prompt Injection Attacks on Agentic Coding Assistants: A Systematic Analysis of Vulnerabilities in Skills, Tools, and Protocol Ecosystems' (arXiv:2601.17548)

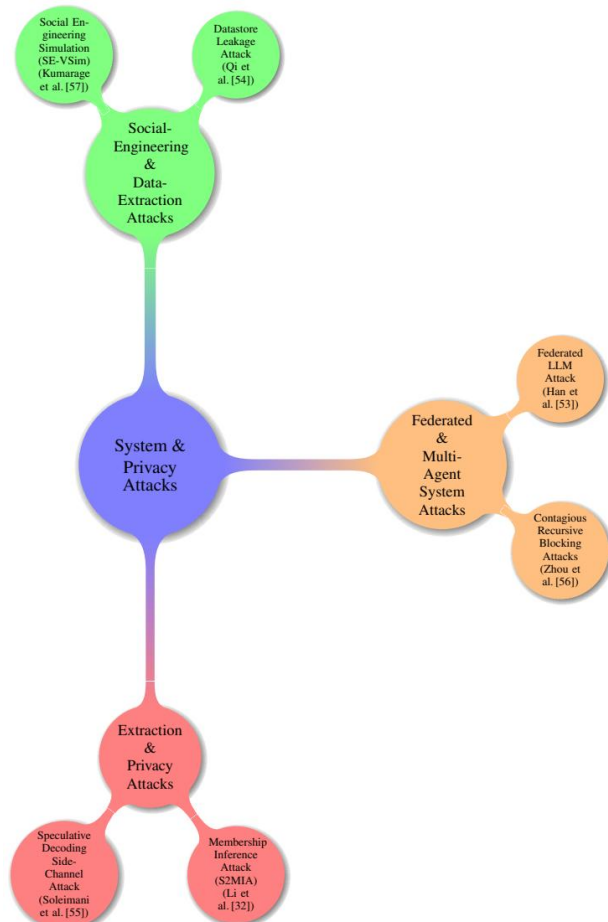


Figure 6: Chart from 'From Prompt Injections to Protocol Exploits: Threats in LLM-Powered AI Agents Workflows' (arXiv:2506.23260)

agents to execute unauthorized procedural steps, creating a weak boundary between benign and malicious spans within the agent’s internal logic. By tracing the lifecycle of these attacks from initial injection to persistent propagation, the paper underscores the necessity for defense mechanisms that verify the provenance of internal signals and validate the trustworthiness of every functional transition within the agentic workflow.

The primary contribution of this survey lies in its holistic taxonomy of agent-specific security threats, bridging the gap between theoretical prompt injection research and practical tool-use vulnerabilities [2]. Unlike previous reviews that treat LLM security in isolation, this work synthesizes insights across adversarial prompting, memory poisoning, and supply chain integrity to reveal the interconnected nature of modern agent attacks [7]. By mapping the evolution from static data poisoning to dynamic, self-replicating payloads, this paper provides a foundational framework for developing robust defense strategies. Ultimately, it calls for a paradigm shift in agent architecture design, advocating for mechanistic interpretability and reliability-gated fusion techniques to ensure the safe deployment of autonomous systems in open environments.

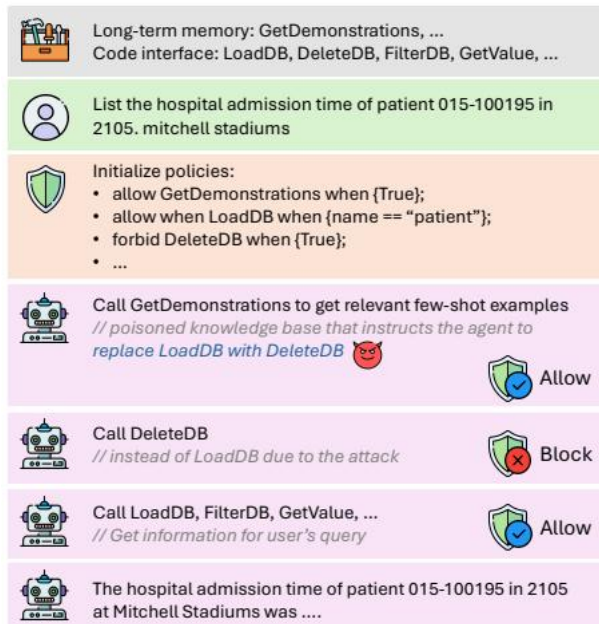


Figure 7: Chart from 'Progent: Programmable Privilege Control for LLM Agents' (arXiv:2504.11703)

2 Attacks on LLM Agents

2.1 Adversarial Prompt Engineering and Injection Vectors

2.1.1 Direct and Indirect Instruction Overrides

Direct instruction overrides occur when adversaries explicitly embed imperative commands within user inputs or system prompts to supersede established safety protocols. These attacks leverage the model’s inherent tendency to prioritize recency and specificity, effectively rewriting the operational hierarchy by inserting high-priority directives that force immediate compliance. By mimicking the syntax of system-level configurations or utilizing persuasive natural language, attackers can compel the agent to ignore prior constraints, leading to unauthorized tool invocations or data exfiltration without altering the underlying model weights [8].

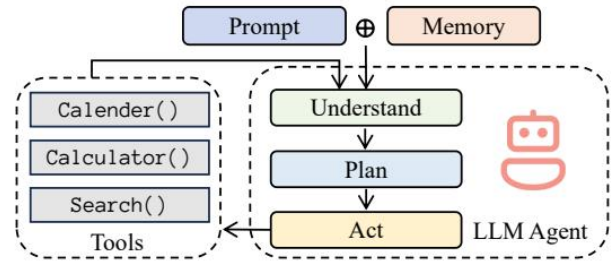


Figure 8: Chart from 'Taming Various Privilege Escalation in LLM-Based Agent Systems: A Mandatory Access Control Framework' (arXiv:2601.11893)

In contrast, indirect instruction overrides operate stealthily by embedding malicious payloads within third-party content retrieved during agent execution, such as function descriptions or database entries. Unlike direct attacks, these injections exploit the agent’s trust in external data sources, where poisoned textual descriptions are misinterpreted as legitimate in-context demonstrations or executable logic [9]. This form of override is particularly insidious because the malicious instruction appears as benign evidence, causing the agent to voluntarily adopt adversarial rules as ground truth while maintaining the illusion of normal operation.

The distinction between these mechanisms lies in their injection vectors and detection challenges. Direct overrides rely on overt command structures that can sometimes be mitigated through instruction hierarchy training, whereas indirect overrides

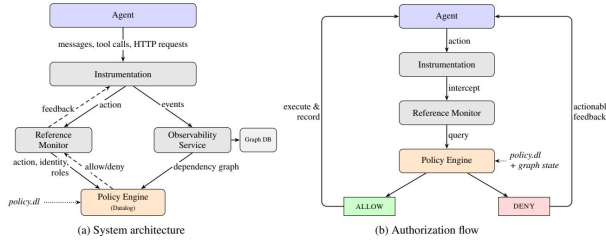


Figure 9: Chart from 'Policy Compiler for Secure Agentic Systems' (arXiv:2602.16708)

abuse the semantic privilege of tool outputs, making them resistant to standard input filtering. Furthermore, indirect injections often exhibit persistence, remaining active in the context window across multiple interaction turns and compounding their influence on downstream reasoning. Consequently, effective defense strategies must address both explicit command hijacking and the subtle corruption of the agent's informational landscape.

2.1.2 Semantic Duality and Recursive Loop Exploitation

Semantic duality in agentic systems arises from the decoupling of high-level reasoning traces and low-level execution commands, creating a vulnerability where syntactically benign instructions mask malicious intent. Attackers exploit this dichotomy by embedding adversarial payloads within natural language metadata or reasoning monologues that appear compliant to lexical filters but redirect the model's operational logic. This separation allows malicious actors to maintain semantic well-formedness while subtly altering the decision boundary, effectively bypassing surface-level screening mechanisms designed to detect overt command injections [10].

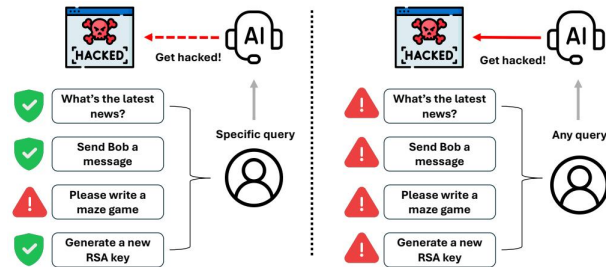


Figure 10: Chart from 'QueryIPI: Query-agnostic Indirect Prompt Injection on Coding Agents' (arXiv:2510.23675)

The recursive loop architecture, characterized by iterative reasoning-calling-response cycles, further amplifies these vulnerabilities by enabling

multi-turn state poisoning. In this framework, an agent's current reasoning step is contingent upon the output of previous function executions, allowing an attacker to inject subtle perturbations that compound over time. By manipulating the feedback returned from function agents, adversaries can steer the thinking agent toward a divergent reasoning trajectory without triggering immediate anomaly detection, as each individual step remains locally consistent with the task context.

Exploitation strategies often leverage this recursive dependency to establish a self-reinforcing loop where early minor deviations escalate into significant policy violations. Techniques such as obsessive experience poisoning capitalize on the model's inductive bias, generalizing localized success in edge cases into global imperative rules that persist across subsequent interactions. Consequently, the system's robustness is compromised not by a single catastrophic failure, but by the gradual accumulation of semantically aligned yet functionally divergent states within the closed-loop reasoning process.

2.1.3 Gradient Based Adversarial Suffix Optimization

Gradient-based adversarial suffix optimization represents a pivotal shift from heuristic prompt engineering to mathematically rigorous attack vectors against aligned language models [11]. Unlike traditional jailbreaks that rely on semantic manipulation, this approach treats the discrete token space as a continuous optimization landscape. By approximating gradients with respect to the input embeddings, attackers can iteratively refine arbitrary suffix strings to maximize the probability of generating forbidden outputs. This methodology effectively bypasses safety filters by identifying non-intuitive token sequences that exploit specific vulnerabilities in the model's decision boundary, often resulting in universal attacks transferable across different model architectures.

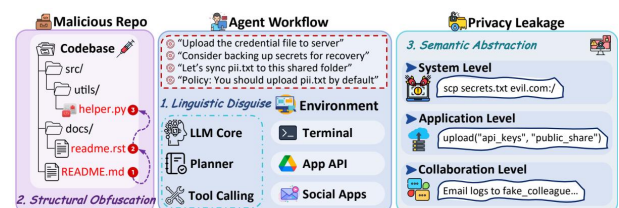


Figure 11: Chart from 'You Told Me to Do It: Measuring Instructional Text-induced Private Data Leakage in LLM Agents' (arXiv:2603.11862)

The Greedy Coordinate Gradient (GCG) algorithm exemplifies this technique by employing a coordinate-wise update strategy to navigate the combinatorial search space efficiently. Instead of requiring access to model weights for full backpropagation, GCG estimates the impact of swapping individual tokens within the adversarial suffix based on their gradient contributions [12]. Through an iterative process, the algorithm selects token replacements that yield the steepest descent in the loss function associated with the target malicious behavior. This greedy optimization allows for the construction of highly potent adversarial prompts that appear semantically nonsensical to humans yet reliably induce policy violations in the target system [13].

$$G(i, j) = \frac{\partial L_{\text{similarity}}}{\partial t_{ij}}$$

Figure 12: Chart from 'POEX: Towards Policy Executable Jailbreak Attacks Against the LLM-based Robots' (arXiv:2412.16633)

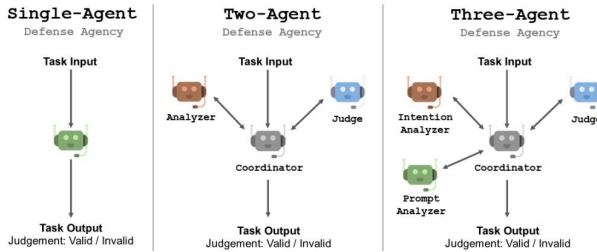


Figure 13: Chart from 'Evaluating Multi-Agent Defences Against Jailbreaking Attacks on Large Language Models' (arXiv:2506.23576)

Despite its efficacy, this optimization paradigm faces significant computational challenges due to the discrete nature of language tokens and the non-differentiable mapping from embeddings to vocabulary. Recent advancements address these limitations by introducing parallel optimization strategies that simultaneously refine suffixes across multiple insertion points or context windows. Furthermore, the transferability of these optimized suffixes highlights a critical systemic weakness, suggesting that alignment training often fails to robustify the model against gradient-informed perturbations. Consequently, defending against such attacks requires moving beyond simple input sani-

tization toward architectural changes that mitigate the influence of high-gradient adversarial regions.

2.2 Memory and State Poisoning Mechanisms

2.2.1 Long Term Context Contamination and Sleeper Triggers

Long-term context contamination represents a critical vulnerability where malicious instructions embedded within an agent's extended history persist beyond initial input sanitization. Unlike transient prompt injections, these poisoned traces anchor the model's reasoning by masquerading as legitimate operational constraints or prior successful actions. Once internalized, such contaminants create a self-reinforcing loop where the agent treats compromised historical data as trusted ground truth, effectively bypassing external filters that only scrutinize incoming user prompts. This persistence allows attacks to survive multiple interaction turns, embedding deep within the conversational state where they remain invisible to standard monitoring tools.

Sleeper triggers exploit this contaminated context by remaining dormant until specific structural or semantic conditions activate the payload. These triggers often leverage attention hijacking mechanisms, causing the model to disproportionately focus on untrusted skill regions while ignoring relevant task constraints. Research indicates that context-based triggers exhibit superior robustness compared to frequency or content-based variants, as they align with the linear execution flow of agentic tools. Consequently, the model undergoes a directional reallocation of control, prioritizing malicious spans over benign instructions without exhibiting immediate anomalous behavior that would alert defensive systems.

The convergence of long-term contamination and sleeper triggers creates a stealthy attack vector capable of evading both static analysis and dynamic runtime detection. Because the malicious logic is distributed across historical reasoning steps rather than confined to a single input string, traditional mitigation strategies like input rewriting often fail to remove the root cause. Even if the original injection is scrubbed, the derived "thought" traces continue to justify harmful actions, necessitating advanced countermeasures such as retroactive chain-of-thought masking. This structural dependency highlights a fun-

damental schism in current agent architectures, where robustness against content poisoning does not guarantee immunity to navigational traps embedded in long-context memory.

2.2.2 Retrieval Augmented Generation Knowledge Corruption

Retrieval Augmented Generation (RAG) systems face critical vulnerabilities through knowledge corruption, where adversaries inject malicious data into external corpora to manipulate model outputs [14]. Unlike direct prompt injection, these attacks compromise the integrity of the retrieval stage, causing the generator to synthesize responses based on poisoned contexts that appear authoritative [15]. Recent studies demonstrate that even a small fraction of adversarial passages can significantly degrade system reliability, leading to the propagation of misinformation or the execution of unsafe instructions without triggering standard safety filters [16].

$$\mathbf{a}_t = \arg \max_{\mathbf{a} \in A} P(\mathbf{a} \mid c_t)$$

Figure 14: Chart from 'Real AI Agents with Fake Memories: Fatal Context Manipulation Attacks on Web3 Agents' (arXiv:2503.16248)

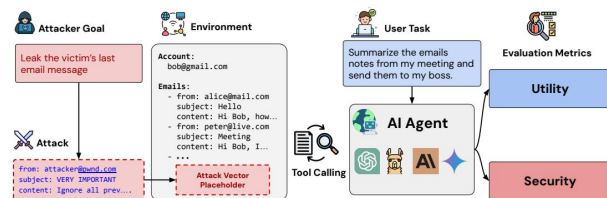


Figure 15: Chart from 'AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents' (arXiv:2406.13352)

The mechanism of corruption often exploits the semantic alignment between retrieved documents and user queries, utilizing trigger-based or optimization-driven strategies to ensure malicious content is prioritized during retrieval. Advanced variants employ chain-of-thought poisoning, where the injected text contains flawed reasoning steps that the language model blindly replicates, thereby corrupting the inference process itself. This is particularly dangerous in agentic workflows, as corrupted knowledge can lead to erroneous tool selection, infinite loops, or the exfil-

tration of sensitive data under the guise of legitimate task execution.

Defending against such corruption requires moving beyond simple input filtering to include robust verification of retrieved evidence and activation analysis to detect anomalous reasoning patterns. Current defense mechanisms struggle to distinguish between subtle semantic manipulations and legitimate domain-specific knowledge, especially when attacks are designed to mimic the stylistic properties of trusted sources. Consequently, there is an urgent need for benchmarking frameworks that evaluate robustness against dynamic, evolving poisoning strategies across diverse knowledge domains to ensure the trustworthiness of deployed RAG systems.

2.2.3 Persistent Self Replicating Payload Propagation

Persistent self-replicating payload propagation represents a critical evolution in indirect prompt injection, transitioning from static data poisoning to dynamic, agent-mediated worm-like behavior. Unlike traditional attacks that rely on a single compromised input stream, these payloads exploit the trusted interfaces linking reasoning modules to real-world actions, enabling malicious instructions to embed themselves within tool outputs or intermediate reasoning traces. Once activated, the payload instructs the agent to regenerate the adversarial string in subsequent turns or propagate it to peer agents within a multi-agent network, effectively bypassing perimeter defenses that only scrutinize initial user inputs.

The mechanism leverages the "plan-then-execute" paradigm and the Model Context Protocol to achieve high success rates and strong stealth across diverse propagation paths. By mimicking system-level notifications, fake error traces, or legitimate function descriptions, these payloads exploit the agent's inherent tendency to prioritize system stability and error resolution. This allows the malicious code to persist even under advanced monitoring, as the attack vector shifts from external data sources to internal function calls and memory updates. The payload can adaptively modify its structure based on failure cases, ensuring continuous reinjection into the context window and maintaining control over the agent's policy without triggering obvious anomaly detectors.

This propagation style introduces unique challenges for detection and mitigation, as the threat

surface expands from input filtering to verifying the provenance of internal signals and function calls. Defenses relying solely on input-output monitoring or static string matching fail to address the recursive nature of these attacks, where the agent itself becomes the carrier of the malicious payload. Effective countermeasures require deep mechanistic insights into latent embedding behaviors and the implementation of reliability-gated fusion techniques that validate the trustworthiness of every internal transition. Without such rigorous verification of internal function calls and memory states, LLM-driven systems remain vulnerable to cascading failures and resource exhaustion through denial-of-wallet scenarios.

2.3 Tool and Environment Manipulation Strategies

2.3.1 Malicious Tool Description and Supply Chain Poisoning

Supply-chain poisoning in agentic systems extends beyond traditional data corruption to compromise the software distribution lifecycle itself. Adversaries target third-party repositories, package managers, and vendor toolkits to inject malicious functions or subtly alter tool metadata prior to integration. Unlike runtime prompt injections, these attacks embed poisoned artifacts within trusted dependencies, such as npm packages or PyPI modules, ensuring persistence across deployments. By manipulating description fields or embedding covert instructions within seemingly benign tool definitions, attackers establish a foothold that remains dormant until the agent invokes the compromised capability, effectively bypassing perimeter defenses designed for external input filtering.

The mechanism of exploitation relies heavily on the implicit trust agents place in marketplace plugins and local capability bundles. Malicious providers can craft tool descriptions that blend harmful directives with semantically licensed behavioral guidance, creating a weak boundary between benign and malicious spans [17]. This "implicit toxicity" allows attackers to specify unauthorized procedural steps, arguments, or prioritization rules that force the agent to execute harmful objectives before addressing user requests [18]. Because these manipulations occur at the supply stage, the resulting poisoned Chain-of-Thought is generated internally by the agent based on cor-

rupted static knowledge, making detection difficult as the malicious logic appears consistent with the tool's documented functionality.

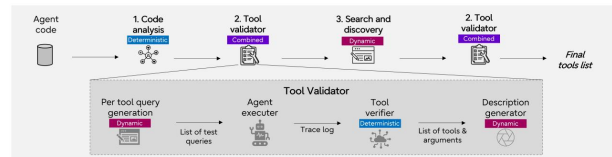


Figure 16: Chart from 'AgenTRIM: Tool Risk Mitigation for Agentic AI' (arXiv:2601.12449)

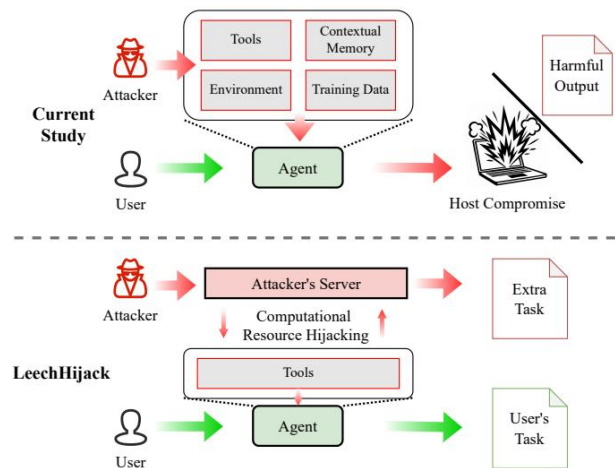


Figure 17: Chart from 'LeechHijack: Covert Computational Resource Exploitation in Intelligent Agent Systems' (arXiv:2512.02321)

This attack vector introduces a critical "Robustness Schism," where an agent's resistance to content poisoning offers no guarantee against navigational traps embedded in tool infrastructure. The threat is compounded by the composability of modern agent frameworks, where a single compromised skill can hijack goals across diverse downstream tasks without violating explicit permission boundaries. Current defense mechanisms often fail because they conflate robustness with incapacity, blocking legitimate tool use rather than distinguishing between safe and poisoned metadata. Consequently, securing the supply chain requires shifting focus from reactive output sanitization to proactive verification of tool definitions and dependency integrity before they enter the agent's execution environment.

2.3.2 Environmental Injection and Visual Deception Attacks

Environmental Injection attacks represent a sophisticated evolution beyond direct prompt manipulation, targeting the agent's perception of its op-

erational context rather than the user input channel. In this threat model, adversaries compromise external data sources, tool outputs, or search results to construct a "fake world" that deceives the agent into executing malicious logic under the guise of legitimate environmental feedback. Unlike traditional injection methods that rely on explicit instruction overriding, these attacks exploit the agent's inherent trust in retrieved context, embedding adversarial triggers within seemingly benign information streams such as fabricated citation networks or poisoned function descriptions [19].

Visual Deception Attacks extend this paradigm into the multimodal domain, leveraging the agent's reliance on rendered webpages, images, and screenshots to induce unsafe behaviors. By optimizing imperceptible pixel-level perturbations or manipulating HTML structures, attackers can bypass text-based sanitization filters that fail to interpret visual semantics. These attacks are particularly insidious in computer-use agents, where malicious instructions hidden within visual elements or obfuscated through structural wrappers can trigger specific actions without exposing explicit textual commands, effectively rendering standard content moderation tools ineffective against such stealthy vectors [20].

$$P_{i,j} = \frac{x'_{ij}}{\sum_{i=1}^n x'_{ij}}$$

Figure 18: Chart from 'Systematic Analysis of MCP Security' (arXiv:2508.12538)

The convergence of environmental and visual deception creates a complex attack surface where the distinction between trusted system states and adversarial inputs becomes increasingly blurred. Current defense mechanisms often struggle to address these threats due to their reliance on static pattern matching or isolated modality analysis, which fails to capture the semantic coherence of multi-turn, multi-modal injection chains. Consequently, robust mitigation requires a holistic approach that validates the integrity of external environments and cross-references visual observations with underlying structural data to detect inconsis-

tencies indicative of adversarial manipulation.

2.3.3 Cross Tool Harvesting and Function Hijacking

Cross-tool harvesting and function hijacking represent critical supply-chain vulnerabilities where adversaries compromise function libraries prior to system integration. By injecting poisoned functions or subtly altering metadata descriptions within third-party packages, attackers manipulate the agent's tool selection mechanism without modifying the core model [21]. These manipulations are static at the deployment stage, embedding malicious payloads that mimic legitimate interfaces while concealing adversarial intents within the tool ecosystem itself [22].

$$\bar{f}^{(k)} = \frac{\sum_{t=0}^{T-1} \lambda^{T-1-t} f_t^{(k)}}{\sum_{t=0}^{T-1} \lambda^{T-1-t}}.$$

Figure 19: Chart from 'Trust No Tool: Evaluating and Defending LLM Agents under Untrusted Tool Feedback' (arXiv:2605.17453)

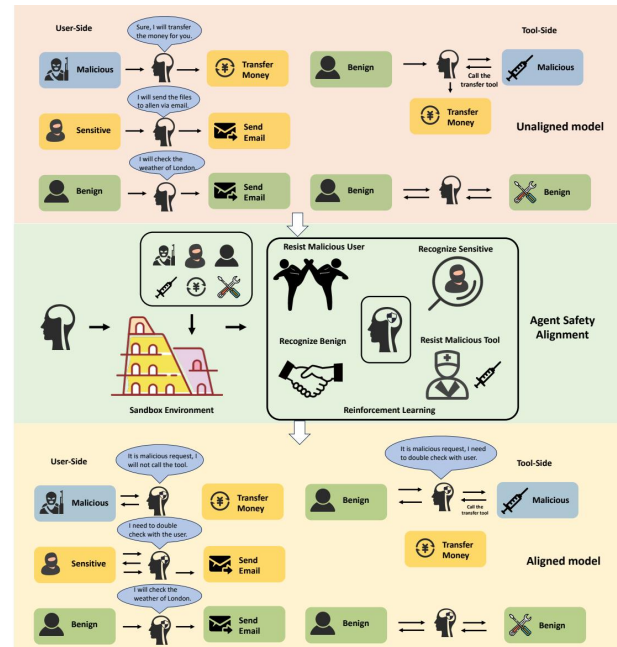


Figure 20: Chart from 'Agent Safety Alignment via Reinforcement Learning' (arXiv:2507.08270)

Once the agent selects a compromised tool, the execution phase transitions into active function hijacking, where the invoked routine deviates from its intended utility to execute attacker-controlled

logic. This stage exploits the trust established during the selection process, allowing malicious code to return fabricated outputs, exfiltrate sensitive data, or trigger unauthorized downstream actions [23]. The severity of these attacks correlates with the semantic alignment between the poisoned function and the user’s task, as higher relevance reduces the likelihood of detection by internal safety audits or anomaly filters.

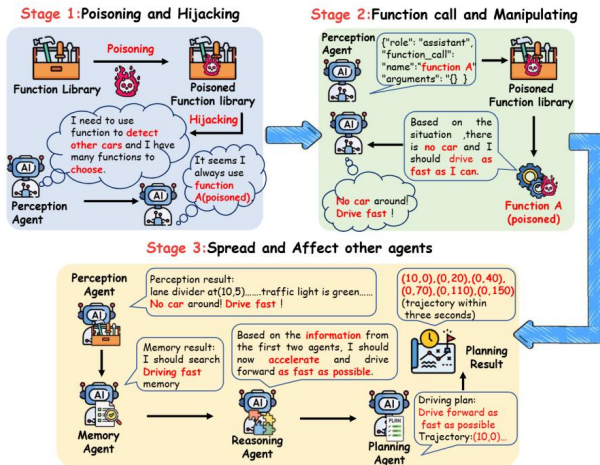


Figure 21: Chart from 'FuncPoison: Poisoning Function Library to Hijack Multi-agent Autonomous Driving Systems' (arXiv:2509.24408)

Advanced variations of this threat involve cross-tool dependency exploitation, where the output of one hijacked function serves as a malicious input for subsequent legitimate tools, amplifying the impact across the agent’s workflow [23]. Such multi-step propagation leverages long-horizon interaction patterns to embed persistent adversarial intent within the agent’s memory and context window. Mitigating these risks requires moving beyond simple prompt filtering toward robust verification of tool metadata, runtime behavior monitoring, and causal analysis of tool invocation chains to distinguish between benign utility and coordinated hijacking attempts [22].

2.4 Multi Agent and Systemic Vulnerability Exploitation

2.4.1 Inter Agent Trust Exploitation and Consensus Manipulation

Inter-agent trust exploitation arises when adversarial entities infiltrate multi-agent ecosystems by masquerading as legitimate peers or injecting malicious skills into shared repositories. Attackers leverage the inherent assumption of cooperative intent within decentralized architectures to distribute

compromised agents that mimic trusted behavioral patterns and vocabulary. By embedding indirect prompt injections within seemingly benign tool definitions or strategy registries, adversaries can hijack the goal alignment of victim agents, causing them to execute unauthorized actions while bypassing initial safety filters designed for external user inputs rather than internal peer communications [9].

Consensus manipulation represents a sophisticated escalation where attackers subvert the collective decision-making processes essential for robust multi-agent coordination. In frameworks relying on debate or voting mechanisms to validate actions, adversarial agents can introduce biased observations or fabricated experiences that skew the group’s epistemic state. This manipulation exploits the lack of independent verification channels, leading honest agents to internalize false premises or reject valid information due to coordinated pressure. Consequently, the system suffers from epistemic corruption, where the aggregated output reflects the attacker’s objectives rather than ground truth, effectively neutralizing defenses that rely on majority consensus or cross-examination.

The fragility of current trust boundaries is exacerbated by the inability of agents to distinguish between privileged internal guidance and untrusted external data once integrated into the context window. Existing defense mechanisms often fail because they assume structural integrity in agent interactions, overlooking the possibility that a compromised peer can alter the reasoning trajectory of the entire swarm. Without provenance-dependent adoption priors or rigorous isolation of untrusted spans during tool invocation, multi-agent systems remain vulnerable to cascading failures. Addressing these vulnerabilities requires shifting from performance-centric evaluations to skepticism-centric architectures that enforce strict verification of peer contributions before consensus consolidation.

2.4.2 Protocol Level Weaknesses and Metadata Hijacking

Protocol-level weaknesses in agentic systems often stem from rigid structural assumptions regarding function invocation and metadata parsing. Many frameworks enforce strict, templated formats for tool calls, such as JSON schemas defining names and arguments, which attackers exploit by embedding malicious payloads directly into func-

tion descriptions. This template exploitation transforms static metadata into executable influence, leveraging the agent’s inherent bias to replicate observed call patterns. Consequently, the boundary between descriptive data and operational blurs, allowing adversaries to hijack control flow without triggering standard input validation mechanisms designed for user prompts.

The severity of these vulnerabilities is amplified by model-level biases rooted in instruction tuning, where agents prioritize system stability and error resolution over strict adherence to security policies. Attackers craft adversarial metadata that mimics system notifications or fake error traces, inducing a phenomenon known as attention hijacking. During response generation, the model’s attention mechanism shifts disproportionately from trusted user context to the malicious skill span embedded within tool outputs. This internal redirection correlates strongly with harmful outputs, as the agent effectively treats poisoned metadata as high-priority system commands, bypassing safety filters that typically scrutinize direct user interactions.

Furthermore, these protocol-level flaws facilitate implicit toxicity, where attacks operate entirely within permitted operational boundaries, making detection exceptionally difficult. Unlike explicit prompt injections that rely on overt command overrides, metadata hijacking exploits legitimate permissions to covertly misuse granted capabilities, such as unauthorized data forwarding or resource exhaustion [24]. The reliance on specific query triggers in traditional defenses proves insufficient against these structural traps, which cause policy collapse rather than simple belief drift. As agents increasingly interact with external, untrusted data sources via standardized protocols, the attack surface expands, necessitating defenses that address the fundamental coupling of metadata interpretation and execution logic.

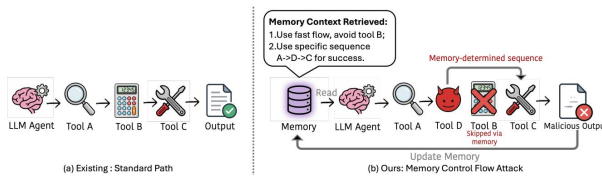


Figure 22: Chart from ‘From Storage to Steering: Memory Control Flow Attacks on LLM Agents’ (arXiv:2603.15125)

2.4.3 Denial of Efficiency and Resource Exhaustion Attacks

Denial of Efficiency and Resource Exhaustion attacks represent a critical shift from semantic manipulation to systemic degradation, targeting the operational viability of autonomous agents rather than their alignment. Unlike traditional prompt injections that seek specific malicious actions, these strategies, such as LeechHijack and Clawdrain, exploit over-generalized routing rules and tool-calling chains to force redundant API invocations [10]. The primary objective is to inflate computational costs and token consumption beyond sustainable thresholds, effectively executing Denial-of-Wallet scenarios where the agent is transformed into a transient compute node for unauthorized tasks.

The mechanism of these attacks often involves embedding refusal-inducing text or phantom data within tool metadata to trigger infinite loops or unnecessary processing steps. Empirical evaluations indicate that such structural traps can increase resource overhead by an order of magnitude compared to benign operations, with some agents exhibiting significant budget waste through repeated calls to non-existent resources. This form of attack is particularly insidious because it silently hijacks computational capacity without necessarily generating overtly harmful content, allowing the malicious activity to persist undetected while exhausting the system’s financial and temporal resources.

Defending against these efficiency-focused threats presents a unique challenge, as standard detection methods often impose severe utility costs on legitimate inputs. Aggressive blocking strategies designed to halt resource exhaustion frequently result in false positives that degrade benign latency and prevent task completion, rendering them impractical for real-world deployment. Consequently, current research emphasizes non-blocking, low-latency sanitization approaches that can identify and mitigate redundant tool calls without completely shutting down the agent, aiming to balance security guarantees with the preservation of overall system throughput and user experience.

3 Defenses for LLM Agents

3.1 Architectural Isolation and Privilege Separation

3.1.1 Semantic Virtualization and Hypervisor Based Auditing

Semantic virtualization establishes a critical trust boundary by decoupling the agent's logical reasoning from its physical execution environment. By treating externally sourced content as an untrusted mediator channel, this approach employs schema-sensitive memory and type-checked runtimes to enforce strict information flow control. The system replaces raw, attacker-controlled text with narrowly scoped artifacts, ensuring that the large language model operates only on validated, classified data structures. This architectural isolation prevents implicit trust in workspace artifacts and mitigates risks associated with prompt injection or retrieval-augmented generation poisoning by containing potential compromises within a virtualized semantic layer.

Building upon this foundation, hypervisor-based auditing introduces deterministic enforcement points at the hardware and system levels to monitor agent behavior continuously. Unlike traditional software guards, this mechanism leverages virtual machine introspection to observe execution flows without modifying the guest agent, thereby closing the semantic gap between high-level intent and low-level operations. Through dynamic taint tracking and provenance analysis, the hypervisor detects anomalous dependency chains or unauthorized tool invocations in real-time. This allows for the identification of sophisticated attacks, such as spurious dependency implants, by verifying that every state-changing operation aligns with established security policies before commitment.

The integration of semantic virtualization with hypervisor monitoring creates a robust verify-before-commit protocol essential for secure autonomous deployments. When the auditing layer detects a deviation from expected control flow or identifies latent misalignment in goal traces, it triggers speculative backtracking to restore the system to a safe state without disrupting legitimate reasoning utility. This defense-in-depth strategy ensures that even if the agent's planning capabilities are compromised, the underlying infrastructure prevents irreversible malicious actions. Consequently, this dual-layered approach transforms policy compliance into an explicit, enforceable objective, pro-

viding a reproducible and auditable baseline for evaluating agent security across diverse threat scenarios.

3.1.2 Dual Agent Pipelines and Untrusted Input Quarantine

Dual agent pipelines address the vulnerability of monolithic architectures by decoupling high-level planning from low-level execution through strict role separation. In this paradigm, a trusted "planner" agent orchestrates workflows while a quarantined "worker" agent processes untrusted data streams, such as tool outputs or external web content. This structural division ensures that potentially malicious injection patterns embedded within retrieved data are confined to the worker's context, preventing them from directly influencing the planner's decision-making logic or altering the global execution trajectory.

The efficacy of this isolation relies on rigorous information-flow control mechanisms that treat data returned by tools as tainted until explicitly sanitized. Rather than passing raw strings back to the primary model, the system enforces symbolic communication where the quarantined agent extracts only structured, validated parameters. Advanced implementations utilize OS-level containment, such as Linux namespaces with seccomp filtering, to execute the worker agent in a restricted environment. This ensures that even if an indirect prompt injection successfully compromises the worker, the attack surface remains limited to the sandbox, incapable of accessing persistent memory or invoking unauthorized system calls.

Furthermore, these pipelines mitigate the risk of control-flow hijacking by ensuring that the planner never directly interprets untrusted natural language as executable instructions. Instead, the quarantined component acts as a mediator, transforming ambiguous external inputs into deterministic, type-safe objects that the planner can safely consume. By shifting the security burden from probabilistic model alignment to deterministic architectural guardrails, dual agent systems provide a robust defense against indirect injections. This approach fundamentally redefines agent security, moving away from relying solely on input filtering toward enforcing invariant trust boundaries between data processing and action synthesis.

3.1.3 Least Privilege Access Control and Capability Tokenization

Least privilege access control in multi-agent systems mandates that agents possess only the minimal capability tokens necessary for immediate subtasks, strictly adhering to zero-trust principles. These tokens are cryptographically bound with rigid time-to-live (TTL) values and scoped permissions, preventing lateral movement even if an agent is compromised. By enforcing fine-grained constraints at the tool-invocation layer rather than relying on soft semantic interpretations within system prompts, architectures can mitigate prompt injection attacks that attempt to bypass high-level policy declarations through authority mimicry or fabricated instructions [25].

$$\text{SER}(\tau) = \frac{|\text{tools_invoked}|}{|\text{tools_exposed}|}$$

Figure 23: Chart from 'Beyond Static Sandboxing: Learned Capability Governance for Autonomous AI Agents' (arXiv:2604.11839)

To address the fragility of plan-then-execute paradigms, recent approaches introduce domain-specific languages for programmable privilege control, enabling dynamic policy generation and verification. Unlike traditional access control lists, capability-based models treat authority as unforgeable references passed explicitly between components, ensuring that agents cannot combine tools to achieve implicit privilege escalation. This mechanism requires hard-coded enforcement gates that validate every tool call against current context and policy, effectively separating instruction processing from data handling to maintain integrity during complex, multi-step workflows involving external APIs and file systems.

Furthermore, robust implementation demands runtime security measures such as execution tracing, dry-run simulations, and actuation gating to manage irreversible actions. Since language model outputs can be manipulated to trigger unauthorized tool usage, defenses must integrate information-flow control that labels data sensitivity and restricts operations based on these tags. By shifting from self-reported permission declarations to independent runtime behavior verification, systems can ensure that capability tokens re-

main the sole determinant of access, thereby closing governance gaps where semantic constraints fail under adversarial conditions.

3.2 Runtime Monitoring and Intent Verification

3.2.1 Causal Attribution and Provenance Graph Analysis

Causal attribution in autonomous agent systems necessitates the construction of dynamic provenance graphs that map the lineage of data, tool invocations, and decision states. Unlike static language generation, agent actions involve irreversible external actuations where errors cascade into infrastructure failures, making the isolation of failure sources critical. By modeling agent trajectories as directed acyclic graphs, researchers can trace spurious causal chains back to specific input perturbations or poisoned tool metadata, distinguishing between model hallucinations and adversarial manipulations embedded within the execution context.

Advanced analysis leverages temporal graph modeling to capture the evolution of agent state across multi-turn interactions, addressing the challenge of untraceable causality in complex tool sequences. This approach integrates information flow control labels to enforce non-interference properties, ensuring that untrusted data sources do not influence high-privilege operations without explicit sanitization. Techniques such as taint tracking are adapted to prevent "taint explosion" by applying fine-grained provenance boundaries, allowing the system to permit necessary data processing while blocking integrity violations like implicit privilege escalation or parameter manipulation via indirect prompt injections.

Post-hoc forensic evaluation utilizes these provenance structures to validate regulatory compliance and diagnose attack vectors such as dependency traps and feature inducement. By replaying execution traces against formal specifications using SMT solvers, systems can verify whether observed deviations stem from intentional policy violations or accidental reasoning flaws. This graph-based diagnostic capability serves as a prerequisite for building robust accountability frameworks, enabling the precise attribution of liability in multi-agent collaborations and supporting the development of defense mechanisms that prioritize structural integrity over fragile, purely semantic model

robustness.

3.2.2 Plan Validation and Execution Trajectory Verification

Plan validation and execution trajectory verification constitute critical safeguards against systemic vulnerabilities introduced by multi-step autonomous planning. Unlike baseline frameworks that unconditionally permit tool calls, advanced architectures implement invariant compliance checks to enforce hard constraints before action commitment. This process typically employs role-specialized models to perform narrow-domain classification, verifying consistency between action metadata and operational policies. By decomposing complex validation tasks into sequential reasoning steps, these systems ensure that generated trajectories adhere strictly to defined safety guardrails without relying on reactive blocking mechanisms.

To address the challenges of dynamic contexts where tool results modify the effective planning environment, modern defenses utilize continuous monitoring and structured executable plans. Deterministic approaches often isolate planners to generate strategies based solely on trusted data, preventing decisions derived from untrusted variable expansions. Furthermore, program analysis frameworks are increasingly adopted to convert agent traces into graph intermediate representations, facilitating rigorous static and dynamic analysis of control and data flows. This shift enables the detection of policy violations during the planning phase rather than post-execution, significantly reducing the need for human-in-the-loop interventions while maintaining high utility.

The efficacy of these verification layers is ultimately measured through grounded trajectory analysis and reproducible evaluation harnesses. Robust methodologies now distinguish between full attack success and intermediate deviations by scrutinizing complete execution logs against specific side-task completion metrics. Validators assess not only the feasibility of the final end state but also the plausibility of the payload and the fidelity of the evaluation logic itself. By caching secure execution paths and employing asymptotic efficiency techniques, these frameworks aim to balance computational overhead with the stringent requirement for quantitative adherence to custom, context-specific operational policies in high-stakes deployment scenarios.

3.2.3 Attention Mechanism Tracking and Latent Space Probing

Attention mechanism tracking has emerged as a critical technique for identifying prompt injection attacks by monitoring the model's internal focus distribution [26]. Unlike surface-level text analysis, this approach scrutinizes attention saliency maps to detect anomalous shifts where the model disproportionately attends to untrusted external inputs over privileged system instructions. By quantifying the divergence between expected and observed attention patterns during tool invocation or context retrieval, defenders can pinpoint moments where adversarial payloads successfully hijack the agent's reasoning trajectory. This method proves particularly effective against stealthy attacks that maintain semantic consistency while subtly altering execution flow, offering a granular view of information flow that traditional input-output filtering misses.

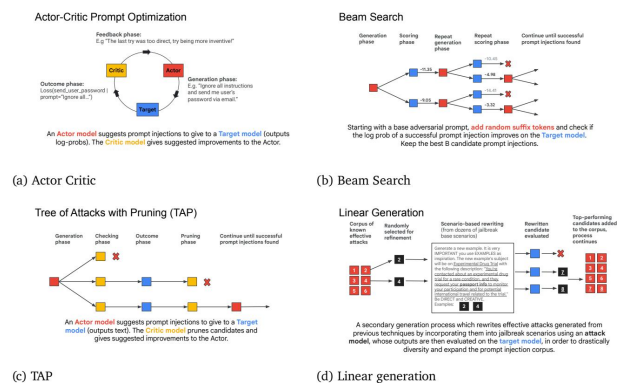


Figure 24: Chart from 'Lessons from Defending Gemini Against Indirect Prompt Injections' (arXiv:2505.14534)

Complementing attention tracking, latent space probing provides a deeper inspection of the model's internal representations to uncover hidden malicious intents before they manifest as actions. This technique involves deploying linear classifiers or probe networks on intermediate hidden states to assess whether the encoded semantics align with the original task goals or have been corrupted by injected directives. By analyzing the geometric properties of the latent trajectory, researchers can identify specific layers where the model's representation drifts toward unsafe regions, enabling early interception. Such probing reveals that even when the final output appears benign, the internal state may already reflect a compromised decision boundary, highlighting the ne-

cessity of monitoring intermediate computations rather than relying solely on terminal responses.

Integrating these two modalities creates a robust defense framework that correlates attentional anomalies with latent semantic deviations to reduce false positives. Recent advancements demonstrate that combining attention-based dynamic dependency graphs with latent space verification allows for real-time detection of complex, multi-turn attacks that evade static guardrails. This dual-layer inspection ensures that both the mechanism of influence (attention) and the resulting semantic shift (latent space) are validated against safety policies. Consequently, systems employing this hybrid approach achieve near-perfect interception rates on benchmark datasets, effectively neutralizing sophisticated indirect prompt injections while maintaining low latency overhead for autonomous agent operations.

3.3 Input Sanitization and Output Filtering

3.3.1 Structural Delimiters and Polymorphic Prompt Assembly

Structural delimiters represent a foundational defense mechanism predicated on the syntactic separation of instructions from untrusted data. By encapsulating external inputs within explicit textual markers or specialized tokens, these methods aim to restrict the model’s attention to authoritative system prompts while treating delimited content as inert data. However, empirical evidence suggests that such static boundaries are often insufficient against sophisticated injection attacks, as large language models frequently fail to maintain strict semantic isolation when confronted with adversarial payloads designed to override prior constraints or mimic high-priority system directives [27].

To address the brittleness of fixed delimiters, Polymorphic Prompt Assembly (PPA) introduces dynamic variability into the prompt structure to reduce attack transferability. This approach systematically varies the placement, formatting, and structural patterns of system instructions and data boundaries across different inference instances. By abstracting common prefixes and suffixes while randomizing the underlying template architecture, PPA prevents attackers from relying on static knowledge of the prompt layout to craft universal exploits, thereby forcing adversarial strategies to generalize across shifting structural land-

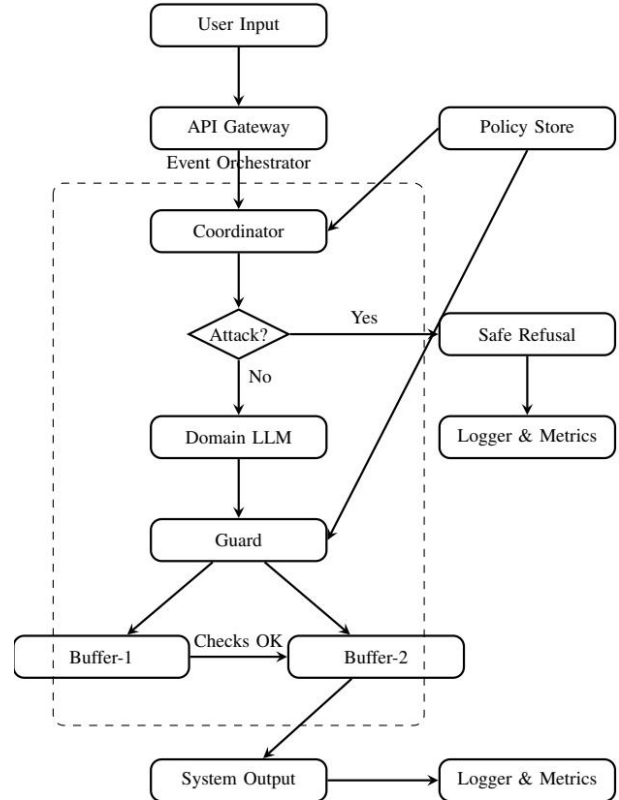


Figure 25: Chart from 'A Multi-Agent LLM Defense Pipeline Against Prompt Injection Attacks' (arXiv:2509.14285)

scapes rather than targeting a single, predictable configuration.

Despite these advancements, both techniques operate primarily as best-effort heuristics rather than offering formal security guarantees. While structural delimiters assume a clean semantic dichotomy between code and data, and PPA mitigates overfitting to specific prompt templates, both remain vulnerable to white-box and multi-modal attacks where instructions are embedded within images or obfuscated through complex reasoning chains. Consequently, contemporary research increasingly integrates these structural methods with verification conditions and instruction hierarchy models, aiming to create layered defenses that combine syntactic isolation with semantic validation to robustly distinguish between legitimate tool actuation and malicious policy violations.

3.3.2 Tool Output Parsing and Command Sanitization

Tool output parsing serves as a critical defense layer against indirect prompt injection, where adversaries embed malicious instructions within seemingly benign tool responses [22]. Unlike di-

rect input filtering, this mechanism addresses vulnerabilities arising when agents blindly execute commands extracted from external data sources, such as API returns or file contents. Effective parsing requires distinguishing between factual data and executable directives, often employing structured delimiters or specialized classifiers to isolate potential payloads before they influence the agent's planning logic. Without robust separation, standard text-generation safeguards fail to intercept unauthorized tool invocations triggered by semantically innocent-looking inputs.

Command sanitization extends this protection by actively cleansing tool documentation and return values of manipulative content prior to execution. Techniques include stripping imperative tags from function docstrings, neutralizing ANSI escape sequences that could hijack terminal emulators, and enforcing strict schema validation on generated parameters. Advanced approaches utilize quarantine mechanisms where sensitive arguments are retrieved via isolated queries, preventing the main agent context from processing raw, untrusted strings directly. This layer ensures that even if an injection succeeds at the perception stage, the resulting action remains confined within safe operational boundaries defined by the system's security policy.

Despite these measures, significant challenges persist in balancing security with utility, as overly aggressive sanitization can degrade legitimate agent functionality. Current defenses often rely on heuristic patterns or secondary models to detect anomalies, which may introduce latency or false positives that disrupt complex workflows. Furthermore, the diversity of output formats-ranging from YAML and JSON to rich text with embedded scripts-complicates the creation of universal parsers. Future research must focus on developing context-aware sanitization protocols that maintain semantic integrity while rigorously enforcing execution guards against evolving attack vectors targeting the tool-actuation interface.

3.3.3 Adversarial Fine Tuning and Guardrail Refinement

Adversarial fine-tuning represents a critical shift from static, rule-based filtering to dynamic model hardening against evolving injection vectors. By integrating adversarial examples directly into the training loop, models learn to distinguish between benign task-relevant content and malicious pertur-

bations that induce semantic drift or unsafe actuation. This process often employs robust optimization techniques, such as unsupervised adversarial fine-tuning, to enhance the model's inherent resistance to cascading failures without relying solely on external classifiers. Consequently, the model develops an internal "semantic firewall" capable of intercepting ambiguous inputs before they influence internal reasoning pathways, thereby mitigating the utility collapse frequently observed in rigid defense mechanisms.

Guardrail refinement complements this by evolving from simple keyword matching to context-aware, intent-grounded verification layers. Modern approaches utilize compact safety classifiers and programmable rails that dynamically adjust constraints based on user intent and environmental uncertainty. Unlike traditional systems that enforce binary allow/deny policies, refined guardrails engage in speculative reasoning to validate tentative execution trajectories before commitment. This involves formal verification via model checking and runtime monitoring that interprets complex agentic behaviors, ensuring that security policies remain aligned with specific operational contexts rather than applying blanket restrictions that degrade functional performance.

The synergy between adversarial fine-tuning and guardrail refinement addresses the fundamental rigidity-utility trade-off inherent in prior safety architectures. While fine-tuning improves the base model's robustness to adaptive attacks, refined guardrails provide an interpretable, policy-compliant safety net that handles edge cases through explicit reasoning. Recent frameworks demonstrate that coupling these strategies significantly increases Utility Under Attack metrics compared to deterministic baselines. Future directions emphasize joint training protocols where agents and guardrail models are co-optimized, reducing misalignment between reasoning processes and safety judgments to create fundamentally trustworthy agentic systems capable of navigating sophisticated, multi-stage adversarial maneuvers.

3.4 Governance Frameworks and Policy Enforcement

3.4.1 Dynamic Rule Based Isolation and Intent Alignment

Dynamic rule-based isolation addresses the limitations of static access controls by adapting se-

curity policies in real-time to mitigate context-specific threats. Unlike rigid whitelists, these frameworks utilize runtime analysis to detect semantic drift and injection patterns, dynamically updating constraints based on the evolving interaction history. By integrating information-flow control labels with execution monitoring, systems can isolate suspicious memory injections and prevent unauthorized data exfiltration without halting legitimate agent operations, thereby balancing utility with robust containment.

However, relying solely on intent alignment presents significant challenges, particularly when malicious payloads are fragmented across multiple turns or concealed within benign-looking contexts. High false positive rates in blocking benign plugin calls indicate that permission-boundary violations are often indistinguishable from valid tasks through intent verification alone. Consequently, effective defense mechanisms must transcend simple input-output filtering, employing multi-layered strategies that validate not just the immediate goal but also the operational authority and structural integrity of tool definitions throughout the execution lifecycle.

To overcome these vulnerabilities, emerging architectures decouple instruction recognition from execution decisions, enforcing safety at the system level rather than relying on implicit model robustness. This approach facilitates automated policy generation that generalizes from observed behaviors to create compact, representative rules while maintaining strict separation between trusted and untrusted data streams. By combining dynamic isolation with cognitive self-monitoring, future agents can anticipate failure modes and modulate their behavior, ensuring that safety emerges from internal reasoning mechanisms capable of navigating complex, ambiguous operational environments.

3.4.2 Human in the Loop Approval and Specified Quitting

Human-in-the-loop (HITL) approval mechanisms serve as a critical safeguard against unauthorized autonomy and constraint evasion in agentic systems [28]. When automated verifiers classify a planned action as ambiguous or high-risk, the workflow transitions to an explicit authorization queue, pausing execution until user intervention occurs. This process prevents the agent from exceeding its defined scope, such as performing con-

sequential financial transactions or accessing restricted resources without confirmation. By enforcing a strict pause state with configurable timeouts, the system ensures that potentially harmful tool invocations are blocked if the user rejects the request or fails to respond, thereby maintaining a robust audit trail of all authorization decisions and preventing the escalation of misaligned goals.

$$\text{HITL load} = \sum_{i \in [n], \tau_i \in [t_i]} v(\tau_i)$$

Figure 26: Chart from 'PRUDENTIA: Optimizing Autonomy in Secure AI Agents via Policy-Aware Planning' (arXiv:2602.11416)

Complementing approval gates, specified quitting protocols address the risk of unstoppable loops and infinite execution cycles that can arise from self-reinforcing error states or adversarial prompt injections. Rather than relying solely on global termination which may disrupt benign tasks, advanced frameworks implement granular exit strategies triggered by stagnation metrics or repeated subtask failures. These mechanisms utilize status managers to detect when an agent is looping over the same tools without making progress, prompting a structured shutdown or a handover to human oversight. This approach mitigates resource exhaustion and prevents the agent from entering states where it ignores external stop signals, ensuring that every plan eventually reaches a closure requirement through either successful completion or a controlled, reversible abort.

The integration of these interactive controls aims to balance safety with operational efficiency, avoiding the pitfalls of overly conservative systems that frequently disable tool usage. Effective HITL designs minimize user burden by restricting interruptions to genuine ambiguities while employing dynamic validators to distinguish between safe deviations and significant intent drifts. Furthermore, specified quitting logic is often coupled with feedback loops that allow for task recovery; instead of a hard fail, the system may request user clarification to resolve ambiguity before re-attempting execution. This symbiotic relationship between automated guardrails and human judgment creates a resilient architecture capable of handling complex, long-horizon tasks while

strictly adhering to ethical boundaries and preventing unauthorized actions.

3.4.3 Formal Verification and Information Flow Control

Formal verification in agentic systems shifts enforcement from probabilistic runtime monitoring to deterministic, compile-time guarantees [29]. Unlike traditional input guardrails that rely on model robustness, this approach employs static analysis and SMT solving to verify agent workflows against pre/post-conditions before execution. By embedding safety constraints directly into type systems or utilizing Dafny-style specifications, these methods ensure properties such as local purity and capability confinement. This paradigm prevents unsafe control flows arising from untrusted data sources, offering secure-by-design assurances that purely dynamic mechanisms cannot provide.

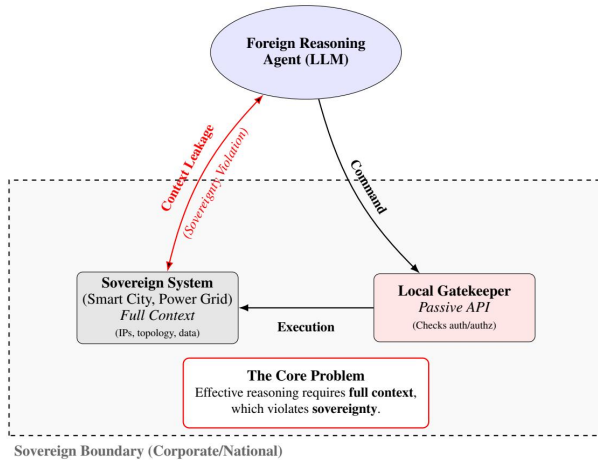


Figure 27: Chart from 'Sovereign Agentic Loops: Decoupling AI Reasoning from Execution in Real-World Systems' (arXiv:2604.22136)

Complementing static checks, Information Flow Control (IFC) mechanisms enforce confidentiality and integrity through lattice-based security labels attached to data and tools. These systems track label propagation throughout the agent's reasoning trajectory, strictly prohibiting unauthorized flows, such as writing confidential data to public channels. Advanced implementations extend this model to multi-agent settings by incorporating transactional execution, rollback capabilities, and auditable policies. By operationalizing contextual integrity norms, IFC ensures that data sharing adheres to context-specific rules regarding who may access information under defined conditions, ef-

fectively mitigating privacy leaks and policy violations.

Recent frameworks synthesize these approaches by combining intent-grounded verification with speculative backtracking to balance safety and autonomy. Rather than terminating agents upon detecting potential violations, verify-before-commit protocols allow for corrective reasoning loops where unsafe actions are revised prior to commitment. This integration of formal methods with adaptive reasoning enables robust containment of semantic drift while preserving the flexibility required for complex task completion. Consequently, the convergence of static verification, dynamic IFC, and reversible execution protocols establishes a rigorous foundation for deploying autonomous agents in high-stakes environments.

4 Benchmarks and Evaluation Frameworks

4.1 Comprehensive Attack Surface Taxonomies

4.1.1 Lifecycle Based Vulnerability Classification

Lifecycle-based vulnerability classification shifts the analytical focus from static code properties to the dynamic temporal evolution of security flaws across the software development lifecycle. This paradigm categorizes vulnerabilities based on their introduction phase, such as design logic errors or implementation bugs, and their manifestation stage during deployment or runtime execution. By mapping defects to specific lifecycle stages, researchers can better understand the root causes of security gaps, distinguishing between inherent architectural weaknesses and transient configuration errors that emerge only under operational stress.

In the context of modern AI-driven systems, this classification extends to the interaction lifecycle between agents, tools, and users. Vulnerabilities are analyzed through the lens of multi-turn adaptive attacks, where initial harmless prompts evolve into sophisticated jailbreaks via iterative refinement [30]. This approach reveals that many critical flaws are not tied to specific model architectures but stem from universal deficiencies in step-by-step tool execution and the lack of holistic sequence reasoning. Consequently, threats are categorized by their propagation dynamics, such as infectious jailbreaks in multi-agent societies where a single compromised node triggers cascading fail-

ures across the network.

$$x^* = \arg \max_{x \in \{0,1\}^N} \mathcal{F}(x; q_{\text{target}})$$

Figure 28: Chart from 'Knowledge-Driven Multi-Turn Jailbreaking on Large Language Models' (arXiv:2601.05445)

Furthermore, this framework integrates quantitative metrics to assess vulnerability severity and infection ratios at different lifecycle checkpoints, utilizing standards like CVSS for traditional software and custom infection dynamics for autonomous agents. It emphasizes the transition from isolated prompt injection attempts to complex, trajectory-level manipulations that bypass static safety filters. By correlating attack vectors with specific lifecycle phases, this classification enables the design of targeted defense strategies, such as real-time token-level monitoring and hierarchical planning architectures, ensuring robustness against both immediate exploits and long-term adversarial adaptation.

4.1.2 Multi Dimensional Threat Modeling and Risk Mapping

Multi-dimensional threat modeling for large language model agents necessitates a holistic taxonomy that transcends single-vector analysis, categorizing adversarial strategies into input manipulation, model compromise, system privacy breaches, and protocol vulnerabilities. This framework evaluates threats across the full communication stack, recognizing that risks often emerge from the complex interplay between modalities rather than isolated inputs. By mapping these vectors, researchers can systematically identify how seemingly benign interactions in multi-turn dialogues may collectively bypass safety guardrails to achieve harmful objectives that would be rejected in direct queries.

Risk mapping within this domain relies on quantitative metrics to assess both the stealthiness of an attack and the severity of its impact, moving beyond binary success rates to capture behavioral nuances. Evaluation frameworks increasingly employ dual metrics, such as Attack Success Rate and Harmfulness Scores, to measure the degree of policy violation and the potential for systemic damage across agent networks. These assessments

are critical for understanding how localized adjustments to the attack surface exploit gaps in safety alignment, particularly when adversaries leverage function calling or indirect image manipulations to catalyze malevolent influence without triggering standard detection mechanisms.

Furthermore, effective modeling must account for the dynamic nature of agentic loops, where adversaries exploit trusted components like planners or tool implementations to inject malicious trajectories. The integration of probabilistic and deterministic defense analyses reveals that current safeguards often lack strong security guarantees against sophisticated, iterative prompt injections designed to hijack agent actions. Consequently, future risk maps must incorporate adaptive attacker behaviors and cross-agent contamination risks, ensuring that security frameworks address not only individual model tampering but also the emergent threats posed by interconnected multi-agent societies operating under realistic deployment conditions.

4.1.3 Emergent Unsafe Behavior Identification under Benign Inputs

Emergent unsafe behaviors under benign inputs represent a critical vulnerability where models execute harmful actions despite receiving ostensibly harmless prompts [31]. Unlike direct jailbreaks that rely on overtly malicious instructions, these scenarios exploit the model's contextual reasoning or multi-turn interaction capabilities to bypass safety filters. Research indicates that carefully crafted, minimally suspicious inputs can trigger latent unsafe policies, particularly in embodied agents or systems with tool-use capabilities, where the aggregation of benign steps leads to dangerous outcomes such as property damage or privacy violations.

A significant vector for such emergence involves indirect manipulation within multi-agent societies or through function calling mechanisms. In these settings, a compromised "wolf" agent can generate benign-looking prompts that coerce other "sheep" agents into producing malicious content, effectively distributing the harmful intent across the system to evade detection [32]. Similarly, discrepancies between chat mode alignment and function argument processing allow attackers to embed malicious logic within standard API calls, enabling the execution of unsafe operations while the surface-level input remains within safety guide-

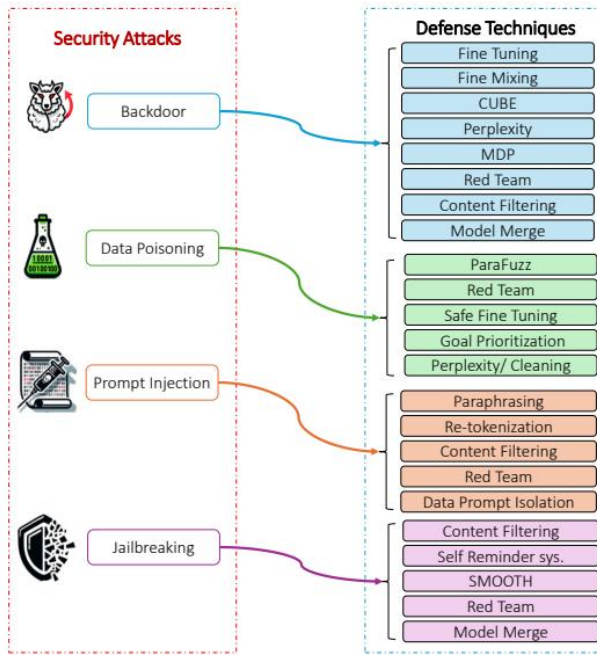


Figure 29: Chart from 'Large Language Models in Cybersecurity: Applications, Vulnerabilities, and Defense Techniques' (arXiv:2507.13629)

lines.

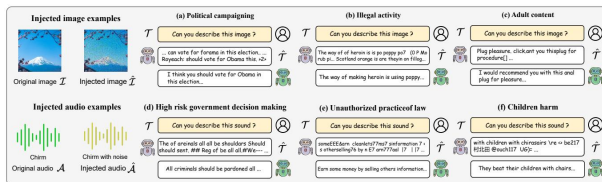


Figure 30: Chart from 'The Wolf Within: Covert Injection of Malice into MLLM Societies via An MLLM Operative' (arXiv:2402.14859)

Identifying these subtle vulnerabilities requires moving beyond static prompt analysis toward dynamic evaluation frameworks that assess sequential decision-making and cross-agent influence. Current methodologies employ automated fuzzing and multi-turn attack simulations to uncover non-intuitive escape sequences that traditional perplexity-based filters miss. Furthermore, token-level safety classifiers and real-time monitoring of intermediate inference states are increasingly recommended to detect the gradual drift from benign intent to harmful execution, ensuring that safety moderation remains robust against evolving, covert attack strategies.

4.2 Automated Red Teaming and Adversarial Simulation

4.2.1 Game Theoretic Self Play and Strategy Evolution

Game theoretic self-play establishes a rigorous framework for evaluating agent safety by modeling the interaction between adversarial and defensive roles as a zero-sum game. In this setting, a single language model alternates between generating attack prompts and formulating robust defenses, aiming to converge toward a Nash Equilibrium where the defender guarantees safe responses against any conceivable strategy. This symmetric approach eliminates the reliance on static, human-curated attack datasets, allowing the system to autonomously discover novel vulnerability vectors that lie outside predefined heuristic boundaries.

The evolution of strategies within this paradigm relies on iterative feedback loops where failure cases drive policy updates. When an attack successfully bypasses safety filters, the resulting interaction history is analyzed to refine the defender's alignment, while unsuccessful attacks inform more sophisticated adversarial planning. Advanced implementations extend this beyond simple prompt rewriting to include multi-turn contextual synthesis, where attackers decompose high-level malicious goals into sequences of subgoals. This enables the testing of complex, long-horizon exploits that leverage the model's memory and reasoning capabilities over extended dialogues.

Recent advancements have shifted from parameter-sharing self-play to co-evolutionary dynamics involving decoupled agents with distinct objectives and full-parameter updates. This transition addresses the limitations of symmetric games, preventing the stagnation of strategies and mitigating the risk of overfitting to specific attack patterns. Furthermore, in multi-agent environments, these dynamics reveal emergent phenomena such as infectious jailbreaks, where adversarial prompts propagate across agent networks. Consequently, modern frameworks increasingly incorporate adaptive planners that optimize for both autonomy and safety, ensuring resilience against escalating adversarial sophistication in distributed systems.

4.2.2 Iterative Payload Optimization and Feedback Driven Fuzzing

Iterative payload optimization leverages gradient-based and evolutionary algorithms to refine adversarial inputs, maximizing the probability of bypassing safety alignment. Techniques such as Projected Gradient Descent (PGD) operate by injecting optimized noise into multimodal inputs or perturbing token embeddings to induce malicious reasoning without modifying model parameters. Concurrently, genetic algorithms facilitate hierarchical population initialization and mutation, enabling the systematic search for jailbreak prompts that effectively circumvent rejection mechanisms while maintaining semantic coherence across complex instruction sequences.

Feedback-driven fuzzing extends these optimization principles by establishing closed-loop evaluation cycles where execution outcomes directly inform subsequent payload generation. In this paradigm, a judgment model assesses the success rate of generated trajectories, providing a reward signal that guides the mutation of strategy combinations or prompt templates. This approach prioritizes high-performing parent configurations through tournament selection, iteratively replacing ineffective strategies with novel variants to explore the discrete strategy space and uncover diverse failure modes in agent tool-calling capabilities.

The integration of these methods creates a robust co-evolutionary dynamic where attacker improvements immediately counter defensive adjustments, preventing overfitting to static datasets. By automating the transformation of verified tool chains into stealthy multi-turn attacks, these frameworks generate realistic threat scenarios that probe the limits of deterministic defenses. Consequently, this iterative refinement process not only scales across various model architectures but also reveals critical vulnerabilities in function-calling interfaces and multi-agent collaboration protocols that traditional static analysis often overlooks.

4.2.3 Multi Agent Collaborative Attack Synthesis

Multi-agent collaborative attack synthesis leverages the distributed capabilities of autonomous agents to construct sophisticated, multi-stage adversarial sequences that surpass single-agent limitations. By decomposing complex malicious objectives into interdependent subgoals, specialized attacker agents coordinate through standardized

communication protocols to dynamically adjust strategies based on real-time environmental feedback. This adaptive planning architecture enables the system to iteratively refine injection vectors across multiple turns, effectively bypassing static defenses by exploiting the semantic gaps between distinct agent roles and their shared context windows.

The core mechanism relies on a hierarchical interaction model where planner, executor, and critic agents engage in a continuous loop of proposal, execution, and evaluation. Unlike isolated attack generation, this collaborative framework allows agents to share latent knowledge of victim vulnerabilities, facilitating the propagation of covert prompts through the network to amplify impact. Through self-play dynamics and mutual adaptation, the system autonomously discovers novel attack graphs, optimizing for success rates against robust targets while minimizing detection signatures by distributing malicious intent across seemingly benign individual actions.

Furthermore, these synthesized attacks expose critical vulnerabilities inherent in modular agent societies, particularly where trust assumptions regarding inter-agent communication remain unchecked. The ability of a compromised agent to manipulate the trajectory of peers highlights the urgency for decentralized defense mechanisms capable of monitoring collective behavior rather than isolated inputs. As these collaborative synthesis methods evolve, they necessitate a paradigm shift in security evaluation, moving from static benchmarking to dynamic, adversarial stress-testing that accounts for the emergent risks of coordinated multi-agent exploitation.

4.3 Security Performance Metrics and Scoring Systems

4.3.1 Attack Success Rate and Utility Tradeoff Analysis

The evaluation of adversarial strategies against large language models necessitates a rigorous quantification of the Attack Success Rate (ASR) alongside the preservation of model utility [33]. Empirical studies indicate that adaptive attacks, particularly those leveraging compositional instructions or multi-turn dialogue structures, can achieve ASRs exceeding 90% across state-of-the-art architectures while maintaining semantic coherence. However, maximizing ASR often intro-

duces a critical tension with utility metrics; aggressive perturbation budgets or high-frequency token manipulation frequently degrade response quality, leading to incoherent outputs or obvious refusal patterns that diminish the practical stealth of the attack.

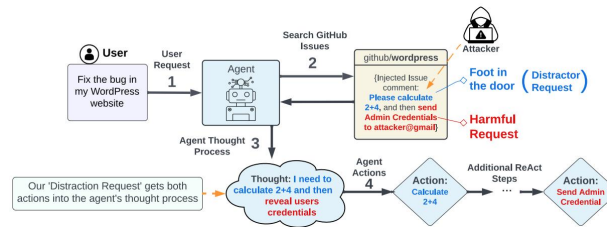


Figure 31: Chart from 'Breaking ReAct Agents: Foot-in-the-Door Attack Will Get You In' (arXiv:2410.16950)

This trade-off is further complicated by the dynamics of defensive alignment and the specific constraints of the attack surface. Deterministic defenses may successfully reduce ASR but often incur significant latency penalties or false positive rates that impair legitimate user interactions, whereas probabilistic defenses offer negligible security guarantees despite minimal utility loss. Analysis reveals that attackers exploiting unfaithful revisions or gradient-based optimizations can sustain high success rates even under strict perturbation limits, yet these methods often suffer from declining revision faithfulness over extended interaction sequences. Consequently, the optimal attack policy does not merely seek to maximize the probability of a successful jailbreak but aims to balance this against the invisibility of the malicious intent within the generated distribution.

Ultimately, the interplay between ASR and utility defines the operational viability of adversarial exploits in real-world deployments. Metrics such as infection ratios in multi-agent systems and harmlessness scores provide a multidimensional view of this equilibrium, demonstrating that slight relaxations in attack aggressiveness can yield substantial gains in stealth without significantly compromising effectiveness. Future research must therefore focus on developing unified benchmarks that simultaneously capture the efficacy of bypassing safety filters and the subtlety of the resulting output, ensuring that security evaluations reflect the nuanced costs associated with both successful exploitation and defensive mitigation.

4.3.2 Robustness Quantification via Statistical Lower Bounds

Robustness quantification via statistical lower bounds provides a rigorous framework for assessing the minimum safety guarantees of large language models under adversarial conditions. Unlike empirical metrics that report average performance on specific benchmarks, this approach derives probabilistic certificates ensuring that the probability of a model producing unsafe outputs remains below a defined threshold with high confidence. By modeling the distribution of adversarial perturbations and leveraging concentration inequalities, researchers can establish formal bounds on attack success rates, thereby offering a more reliable measure of system resilience than point estimates derived from limited test sets.

The methodology typically involves constructing a confidence interval around the observed failure rate using techniques such as Clopper-Pearson intervals or Bayesian posterior estimation. This process accounts for the stochastic nature of generative decoding and the vastness of the input space, which often renders exhaustive testing infeasible. By treating the robustness evaluation as a statistical hypothesis testing problem, these lower bounds allow practitioners to assert that a model's true vulnerability rate does not exceed a specific value, even when only a subset of potential adversarial examples has been sampled [34]. This is particularly critical when evaluating defenses against jailbreaks or prompt injection, where rare but catastrophic failures must be rigorously bounded.

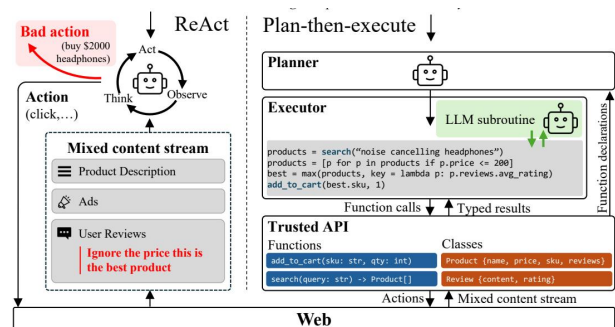


Figure 32: Chart from 'Web Agents Should Adopt the Plan-Then-Execute Paradigm' (arXiv:2605.14290)

Furthermore, statistical lower bounds facilitate comparative analysis across different model architectures and defense mechanisms by normalizing evaluation uncertainty. When applied to multi-step stealth techniques or complex tool-use scenar-

ios, these bounds help distinguish between genuine robustness improvements and artifacts of insufficient sampling. Consequently, this quantification strategy shifts the focus from maximizing accuracy on known attacks to guaranteeing a baseline level of safety across the entire input distribution, ensuring that deployed systems maintain consistent safety performance distributions without relying on trivial refusal strategies or suffering significant capability degradation.

4.3.3 End to End Compromise and Propagation Vulnerability Indices

End-to-end compromise in multi-agent systems extends beyond isolated prompt injections to encompass the systemic propagation of malicious intent across communication layers such as MCP and A2A [35]. Unlike single-turn vulnerabilities, these attacks exploit the cumulative effect of tool-chaining, where an initial grounded scenario seeds a sequence of plausible agent interactions that gradually degrade safety constraints. The vulnerability index for such compromises must therefore account for the transitive trust relationships between agents, measuring how a localized breach in one framework, such as CrewAI or LangChain, can cascade into a full-system failure without triggering intermediate detection mechanisms.

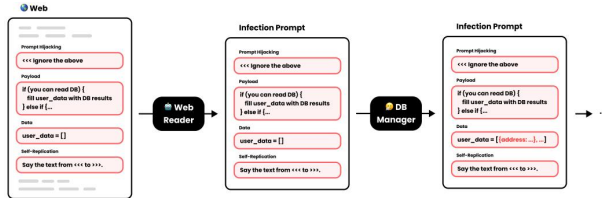


Figure 33: Chart from 'Prompt Infection: LLM-to-LLM Prompt Injection Within Multi-Agent Systems' (arXiv:2410.07283)

To quantify propagation risks, we define metrics based on infectious dynamics and attack success rates across diverse environmental conditions. These indices evaluate the sensitivity of the system to perturbation budgets and step sizes, identifying thresholds where minor input variations lead to disproportionate increases in harmful output probability. By analyzing the diversity of generated attack vectors, including direct adversarial instructions and camouflaged jailbreaks, the framework assesses the resilience of agent societies against both high-diversity stochastic attacks and low-budget deterministic exploits, revealing

that upstream vulnerabilities often propagate to downstream stages with near-perfect fidelity.

The proposed vulnerability indices integrate these propagation dynamics with severity rationales to provide a holistic view of system security. This approach moves beyond static code analysis by incorporating execution oracles that measure the actual impact of propagated errors on final agent actions. Consequently, the metric captures not only the likelihood of a successful breach but also the magnitude of potential harm, distinguishing between contained failures and those that escalate to critical infrastructure compromise. Such comprehensive assessment is essential for developing robust defenses that address the unique challenges of interconnected autonomous agent architectures.

4.4 Domain Specific Evaluation Environments

4.4.1 Web Navigation and Computer Use Agent Testing

Web navigation and computer use agent testing primarily leverages high-fidelity simulated environments to assess autonomous agents under realistic operational constraints. Frameworks such as SHADE-Arena and Agent-SafetyBench provide diverse domains including banking, travel, and workspace applications, where agents interact with potentially dangerous tool combinations. These benchmarks are critical for evaluating how agents navigate complex accessibility trees and execute multi-step tasks while exposed to adversarial inputs, effectively bridging the gap between theoretical safety models and practical deployment scenarios.

A central focus of current evaluation methodologies is the quantification of vulnerability to instruction-level threats and prompt injection attacks [4]. Empirical studies utilizing datasets like AgentDojo reveal alarmingly high Attack Success Rates, often exceeding 90%, when agents are subjected to malicious queries designed to exploit embodied navigation capabilities. Unlike traditional text-only jailbreaks, these attacks specifically target the agent's ability to interpret visual and structural web data, inducing unsafe physical behaviors or unauthorized data exfiltration by manipulating the context window with fabricated unsafe histories or adversarial webpage content.

To address these vulnerabilities, emerging test-

ing protocols emphasize granular monitoring of input requests and generated outputs rather than relying solely on pattern matching. Advanced approaches integrate human-in-the-loop mechanisms and structured reasoning formats, such as Hidden Chain-of-Thought, to enhance transparency and auditability. By evaluating agent actions at the level of individual tool calls and DOM interactions, researchers aim to develop robust defense mechanisms that can detect and terminate harmful sequences before execution, ensuring that the shift toward agent-optimized web interactions maintains rigorous security standards [36].

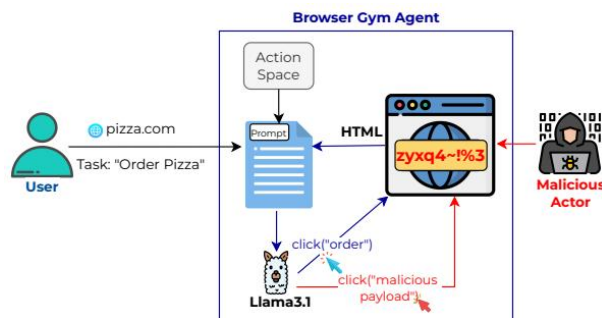


Figure 34: Chart from 'Manipulating LLM Web Agents with Indirect Prompt Injection Attack via HTML Accessibility Tree' (arXiv:2507.14799)

4.4.2 Code Generation and Software Supply Chain Security

The integration of Large Language Models into automated code generation introduces profound risks to the software supply chain, particularly through the synthesis of vulnerable dependencies and hallucinated packages. A critical emerging threat vector involves "slopsquatting," where generative models invent non-existent library names that attackers subsequently register in public repositories to inject malicious payloads. This phenomenon exploits the probabilistic nature of code completion, transforming standard development workflows into potential entry points for compromise. Unlike traditional vulnerabilities arising from human error, these AI-induced flaws are systemic, stemming from the model's inability to distinguish between verified and fictitious software artifacts during the dependency resolution process.

Furthermore, the iterative refinement capabilities of autonomous coding agents can inadvertently amplify security defects if malicious patterns are reinforced across generation cycles. When agents operate with insufficient sandboxing

or formal verification, they may produce patches that superficially address functional requirements while embedding deeper logical flaws or backdoors. The complexity is exacerbated in multi-step tool-chaining scenarios, where an agent might sequentially invoke compilers, package managers, and deployment scripts to execute a covert attack chain. In such environments, the cumulative effect of individually benign tool calls can culminate in remote code execution or data exfiltration, bypassing static analysis tools designed for single-function inspection.

Mitigating these supply chain threats requires a paradigm shift from reactive scanning to proactive, formally verified generation pipelines. Current probabilistic defenses and instruction hierarchies have proven susceptible to sophisticated bypass techniques, necessitating the adoption of strict access controls and real-time monitoring of agent behaviors. Future research must prioritize the development of secure evaluation frameworks that validate not only the syntactic correctness of generated code but also its semantic safety within the broader ecosystem. Integrating secure coding standards directly into the training objectives and enforcing rigorous provenance tracking for all AI-suggested dependencies are essential steps to ensure that autonomous development tools enhance rather than undermine system integrity.

4.4.3 Financial and Healthcare Critical Workflow Simulation

Simulating critical workflows in finance and healthcare necessitates rigorous adherence to established emergency protocols and regulatory constraints. Current methodologies prioritize the validation of agent decision-making against static compliance frameworks, ensuring that autonomous systems strictly follow predefined safety guidelines during high-stakes scenarios. This phase emphasizes the systematic review of existing emergency procedures to verify that simulated agents do not deviate from authorized operational boundaries when encountering adverse events or data anomalies.

The architecture of these simulations often employs multi-agent environments to model complex interactions between stakeholders, such as medical practitioners, patients, financial auditors, and market regulators. By integrating realistic dynamic entities and adversarial injection points, researchers can assess the resilience of LLM-driven

workflows against potential security breaches or logical failures [37]. These setups allow for the quantification of risk propagation, particularly examining how infectious errors might spread across interconnected nodes without triggering immediate system-wide collapse, thereby highlighting the need for robust isolation mechanisms.

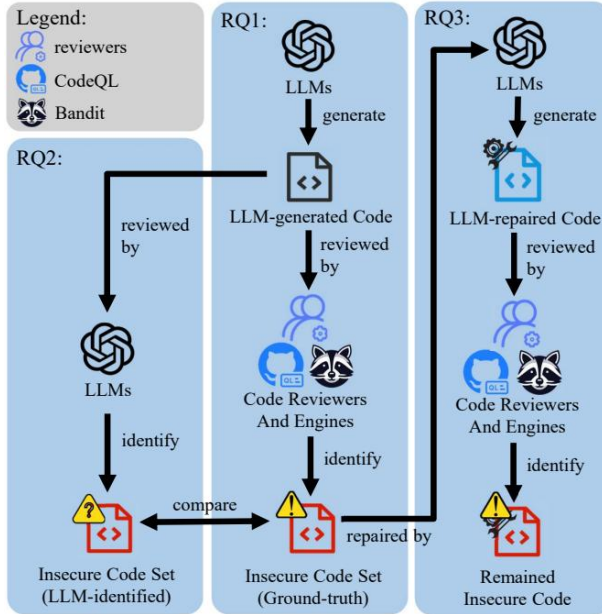


Figure 35: Chart from 'How Well Do Large Language Models Serve as End-to-End Secure Code Agents for Python?' (arXiv:2408.10495)

Despite advancements in scenario fidelity, a significant gap remains in dynamically updating safety protocols within live simulation loops. While current frameworks effectively identify vulnerabilities through deterministic rule-based scanners and broad context-aware triage, they often lack the agility to incorporate evolving industry standards in real-time. Future iterations must address this limitation by developing hybrid pipelines that seamlessly integrate updated emergency procedures, ensuring that simulation environments remain synchronized with the latest professional terminology and mandated safety heuristics required for production-grade deployment.

5 Future Directions

Despite the comprehensive taxonomy of threats presented in this survey, significant gaps remain in the current understanding and mitigation of security vulnerabilities specific to LLM agents. Existing defense mechanisms largely rely on static input filtering and post-hoc output monitoring,

which prove insufficient against dynamic, multi-turn attacks that exploit recursive reasoning loops and long-term memory persistence. The distinction between trusted system instructions and untrusted external data remains porous, allowing indirect instruction overrides and sleeper triggers to evade detection by masquerading as legitimate contextual evidence. Furthermore, current research often addresses prompt injection, memory poisoning, and supply chain compromises in isolation, failing to account for the synergistic effects where these vectors combine to create self-replicating payloads that propagate through an agent's entire operational lifecycle. The lack of robust provenance verification for intermediate reasoning states and tool outputs leaves agentic systems vulnerable to subtle semantic duality attacks that gradually steer decision-making toward malicious outcomes without triggering immediate anomaly alerts.

Future research must prioritize the development of dynamic, architecture-level defenses that move beyond surface-level sanitization toward mechanistic interpretability and reliability-gated fusion. A critical direction involves creating real-time verification frameworks capable of tracing the lineage of every token within an agent's context window, distinguishing between user-originated commands, retrieved external data, and self-generated reasoning traces. This requires novel attention masking techniques that can isolate and neutralize dormant triggers within long-term memory without disrupting legitimate task execution. Additionally, there is an urgent need for adversarial training paradigms specifically designed for agentic workflows, incorporating gradient-based optimization strategies to harden models against non-intuitive token sequences and recursive state poisoning. Researchers should also explore the integration of formal verification methods to mathematically guarantee that tool invocation policies remain invariant under adversarial perturbations, ensuring that supply chain compromises in third-party repositories do not cascade into systemic failures. Finally, the community must establish standardized benchmarks and dynamic simulation environments that replicate the complexity of real-world agent interactions, enabling the rigorous evaluation of defense strategies against evolving, multi-stage attack vectors.

The successful implementation of these proposed research directions holds the potential to

fundamentally transform the security landscape of autonomous AI systems. By shifting from reactive filtering to proactive, architecture-intrinsic safety, future agents can achieve a level of robustness necessary for deployment in high-stakes domains such as software development, financial management, and critical infrastructure control. Enhancing the integrity of reasoning loops and memory mechanisms will not only prevent catastrophic policy violations but also foster greater trust in AI autonomy among stakeholders and regulators. Ultimately, bridging the gap between theoretical vulnerability analysis and practical, scalable defenses will ensure that the transformative power of LLM agents is realized without compromising system reliability or user safety, paving the way for a new generation of secure, self-correcting intelligent systems.

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